Quantum Operations

Chapter 4

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Outline

4.1 Unitary Evolution and Measurement Updates
Closed-system evolution and Hamiltonians
Evolution of density operators
Measurement probabilities and output states
Unread and successive measurements

Unitary Evolution and Measurement Updates

Evolution of a closed system

The evolution postulate

Between two fixed times, a closed quantum system evolves by a unitary operator U on its Hilbert space:

$$\boxed{|\psi\rangle \longmapsto U|\psi\rangle, \qquad U^\dagger U = UU^\dagger = I.}$$

The operator depends on the physical evolution and the two times, but not on the initial state.

Inner products and reversibility

For any $|\psi\rangle, |\varphi\rangle$,

$$\langle\psi|U^\dagger U|\varphi\rangle = \langle\psi \mid \varphi\rangle.$$

Thus evolution preserves normalization and orthogonality. Applying $U^\dagger$ restores the initial state.

Examples: the Pauli operators

The bit flip

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad X|0\rangle = |1\rangle, \qquad X|1\rangle = |0\rangle.$$

Hence $X(\alpha|0\rangle + \beta|1\rangle) = \beta|0\rangle + \alpha|1\rangle$.

The phase flip

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Z|0\rangle = |0\rangle, \qquad Z|1\rangle = -|1\rangle.$$

The computational-basis states keep the same density operators. A superposition can change:

$$Z(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle - \beta|1\rangle, \qquad Z|+\rangle = |-\rangle.$$

Both operators are unitary, with $X^2 = Z^2 = I$.

A classical Hamiltonian: a mass on a spring

A mass $m > 0$ on an ideal spring with constant $k > 0$ satisfies Newton's equation $m\,d^2x/dt^2 = -kx$, where x is the displacement from equilibrium.

For this system, the Hamiltonian is the total energy.

Set $p = m\,dx/dt$. The classical Hamiltonian is the function

$$H(x,p) = \underbrace{\frac{p^2}{2m}}_{\text{kinetic energy}} + \underbrace{\frac{kx^2}{2}}_{\text{potential energy}}.$$

Hamilton's equations

$$\frac{dx}{dt} = \frac{\partial H}{\partial p} = \frac{p}{m}, \qquad \frac{dp}{dt} = -\frac{\partial H}{\partial x} = -kx.$$

Differentiating $dx/dt = p/m$ recovers Newton's equation:

$$\boxed{m\frac{d^2x}{dt^2} = \frac{dp}{dt} = -kx.}$$

Continuous unitary evolution

Assume $U(t)$ is differentiable and unitary, with $U(0) = I$ and $U(t+s) = U(t)U(s)$. This composition law expresses time-homogeneous evolution.

Set $G = \left.\frac{dU(t)}{dt}\right|_{t=0}$. Differentiating $U(t)^\dagger U(t) = I$ at $t = 0$ gives $G^\dagger + G = 0$. For $\hbar > 0$, define $H = i\hbar G$. Then

$$H^\dagger = -i\hbar G^\dagger = i\hbar G = H.$$

Differentiating $U(s+t) = U(s)U(t)$ with respect to s at $s = 0$ gives

$$\frac{dU(t)}{dt} = GU(t) = -\frac{i}{\hbar} HU(t).$$

Thus $|\psi(t)\rangle = U(t)|\psi(0)\rangle$ satisfies

$$\boxed{i\hbar \frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle.}$$

The Schrödinger equation

In continuous time, the state vector satisfies

$$\boxed{i\hbar \frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle.}$$

The Hamiltonian H is Hermitian, and $\hbar > 0$ is a physical constant. We assume that H is independent of time. Here H denotes the Hamiltonian, not the Hadamard operator.

The energy eigenbasis

By the spectral theorem,

$$H = \sum_j E_j |e_j\rangle\langle e_j|, \qquad E_j \in \mathbb{R}.$$

The eigenvalues E_j are the possible energies. Define

$$\boxed{U(t) = e^{-itH/\hbar} = \sum_j e^{-itE_j/\hbar} |e_j\rangle\langle e_j|.}$$

The Hamiltonian generates unitary evolution

Each eigenvalue of $U(t)$ has modulus one, so

$$U(t)^\dagger U(t) = \sum_j e^{itE_j/\hbar} e^{-itE_j/\hbar} |e_j\rangle\langle e_j| = I.$$

Differentiating the finite spectral sum gives

$$\frac{dU(t)}{dt} = \sum_j \frac{-iE_j}{\hbar} e^{-itE_j/\hbar} |e_j\rangle\langle e_j| = -\frac{i}{\hbar} HU(t).$$

Therefore $|\psi(t)\rangle = U(t)|\psi(0)\rangle$ satisfies

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = HU(t)|\psi(0)\rangle = H|\psi(t)\rangle.$$

Also $U(0) = I$, so the initial condition is correct.

For a time-dependent Hamiltonian $H(t)$, the simple exponential formula above generally no longer applies. □

Energy eigenstates and relative phases

An energy eigenvector changes only by an overall phase:

$$U(t)|e_j\rangle = e^{-itE_j/\hbar}|e_j\rangle.$$

Its density operator is stationary:

$$U(t)|e_j\rangle\langle e_j|U(t)^\dagger = |e_j\rangle\langle e_j|.$$

For a superposition $|\psi(0)\rangle = \sum_j c_j|e_j\rangle$,

$$|\psi(t)\rangle = \sum_j c_j e^{-itE_j/\hbar}|e_j\rangle.$$

The energy probabilities $|c_j|^2$ stay fixed, but the relative phase between terms j, k changes by

$$e^{-it(E_j-E_k)/\hbar}.$$

Other measurements can detect this change.

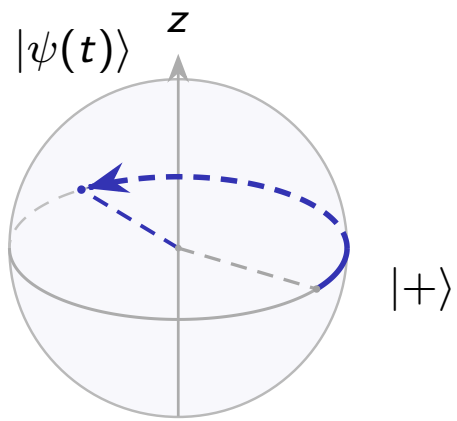
A qubit Hamiltonian

Let $H = \hbar\omega Z/2$, where $\omega > 0$. Then

$$U(t) = \begin{pmatrix} e^{-i\omega t/2} & 0 \\ 0 & e^{i\omega t/2} \end{pmatrix}.$$

Starting from $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, we obtain

$$|\psi(t)\rangle = \frac{e^{-i\omega t/2}|0\rangle + e^{i\omega t/2}|1\rangle}{\sqrt{2}}.$$



Rotation by ωt

Its density matrix and Bloch vector are

$$\rho(t) = \frac{1}{2}\begin{pmatrix} 1 & e^{-i\omega t} \\ e^{i\omega t} & 1 \end{pmatrix}, \qquad \vec{r}(t) = (\cos(\omega t), \sin(\omega t), 0).$$

A computational-basis measurement gives $\mathbb{P}(0) = \mathbb{P}(1) = 1/2$ at every time. The off-diagonal entries still change.

Detecting the relative phase

Measure the evolving state in the X basis $\{|+\rangle, |-\rangle\}$. The amplitude of outcome $+$ is

$$\begin{aligned}\langle + \mid \psi(t)\rangle &= \frac{1}{2}(e^{-i\omega t/2} + e^{i\omega t/2}) \\ &= \cos(\omega t/2).\end{aligned}$$

Thus

$$\boxed{\mathbb{P}(+) = \cos^2(\omega t/2), \qquad \mathbb{P}(-) = \sin^2(\omega t/2).}$$

In particular,

$$\begin{aligned} t = 0: &\quad |\psi(t)\rangle = |+\rangle, \\ t = \pi/\omega: &\quad |\psi(t)\rangle = -i|-\rangle. \end{aligned}$$

The X-basis probabilities reveal the evolution even though the computational-basis probabilities stay fixed.

Every unitary is an exponential

Claim. Every unitary U has the form $U = e^{iK}$ for a Hermitian operator K.

Proof. A unitary is normal. The spectral theorem gives

$$U = \sum_j e^{i\theta_j} |u_j\rangle\langle u_j|, \qquad \theta_j \in \mathbb{R},$$

in an orthonormal eigenbasis. Set

$$K = \sum_j \theta_j |u_j\rangle\langle u_j|.$$

Then $K^\dagger = K$ and $e^{iK} = U$. □

The generator is not unique

Replacing any θ_j by $\theta_j + 2\pi n_j$, $n_j \in \mathbb{Z}$, gives another possible K.
For a fixed duration $\tau > 0$, $H = -\hbar K/\tau$ gives $U = e^{-i\tau H/\hbar}$.

Unitary evolution of a density operator

Suppose an ensemble prepares $|\psi_j\rangle$ with probability p_j:

$$\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|.$$

The same unitary acts on every member of the ensemble.
Therefore

$$\begin{aligned}\rho' &= \sum_j p_j U|\psi_j\rangle\langle\psi_j|U^\dagger \\ &= U\left(\sum_j p_j|\psi_j\rangle\langle\psi_j|\right)U^\dagger.\end{aligned}$$

Hence

$$\boxed{\rho \longmapsto U\rho U^\dagger.}$$

The output depends only on ρ, not on its ensemble decomposition.

Unitary evolution preserves the spectrum

If $\rho|v_j\rangle = \lambda_j|v_j\rangle$, then

$$(U\rho U^\dagger)U|v_j\rangle = U\rho|v_j\rangle = \lambda_j U|v_j\rangle.$$

The vectors $U|v_j\rangle$ form an orthonormal eigenbasis with the same eigenvalues λ_j.

Consequences

Trace and positivity are preserved, so $U\rho U^\dagger$ is a state.

Pure states remain pure

If $\rho = |\psi\rangle\langle\psi|$, set $|\varphi\rangle = U|\psi\rangle$. Then

$$\rho' = U|\psi\rangle\langle\psi|U^\dagger = |\varphi\rangle\langle\varphi|.$$

Since U preserves norms, $|\varphi\rangle$ is normalized, so ρ' is pure.

A mixed state cannot become pure: applying the inverse update $\rho = U^\dagger\rho' U$ to a pure output would give a pure input.

The von Neumann equation

For a fixed Hamiltonian, let $\rho(t) = U(t)\rho(0)U(t)^\dagger$. We have

$$\frac{dU(t)}{dt} = -\frac{i}{\hbar}HU(t), \qquad \frac{dU(t)^\dagger}{dt} = \frac{i}{\hbar}U(t)^\dagger H.$$

The product rule gives

$$\begin{aligned}\frac{d\rho(t)}{dt} &= \frac{dU(t)}{dt}\rho(0)U(t)^\dagger + U(t)\rho(0)\frac{dU(t)^\dagger}{dt} \\ &= -\frac{i}{\hbar}H\rho(t) + \frac{i}{\hbar}\rho(t)H.\end{aligned}$$

Thus the density operator satisfies the von Neumann equation

$$\boxed{\frac{d\rho}{dt} = -\frac{i}{\hbar}[H,\rho], \qquad [H,\rho] = H\rho - \rho H.}$$

Any state commuting with H is stationary.

Measurement operators

A POVM specifies outcome probabilities. To describe the state afterwards, we need additional information. Here we use one measurement operator per recorded outcome.

The measurement postulate

Let $M_x : \mathcal{H} \to \mathcal{K}$ be linear operators indexed by a finite outcome set Ω, satisfying

$$\boxed{\sum_x M_x^\dagger M_x = I_{\mathcal{H}}.}$$

On input ρ, the probability of outcome x is

$$p(x) = \mathrm{tr}[M_x \rho M_x^\dagger].$$

If $p(x) > 0$, the conditional output state is

$$\boxed{\rho_x = \frac{M_x \rho M_x^\dagger}{p(x)}.}$$

The associated POVM

Cyclicity of the trace gives

$$p(x) = \mathrm{tr}[M_x^\dagger M_x \rho] = \mathrm{tr}[\mu(x)\rho], \qquad \boxed{\mu(x) = M_x^\dagger M_x.}$$

Each $\mu(x)$ is positive, since

$$\langle\psi|\mu(x)|\psi\rangle = \|M_x|\psi\rangle\|^2 \geq 0,$$

and the completeness condition gives $\sum_x \mu(x) = I_{\mathcal{H}}$. Thus these operators form a POVM.

Valid probabilities and states

Since $M_x \rho M_x^\dagger \geq 0$, we have $p(x) \geq 0$, and

$$\sum_x p(x) = \mathrm{tr}\left[\left(\sum_x M_x^\dagger M_x\right)\rho\right] = 1.$$

For $p(x) > 0$, the state ρ_x is positive and has trace one.

The conditional state for a pure input

If $\rho = |\psi\rangle\langle\psi|$, then

$$M_x \rho M_x^\dagger = M_x|\psi\rangle\langle\psi|M_x^\dagger, \qquad p(x) = \|M_x|\psi\rangle\|^2.$$

For $p(x) > 0$, the conditional output vector is therefore

$$\boxed{|\psi_x\rangle = \frac{M_x|\psi\rangle}{\|M_x|\psi\rangle\|}, \qquad \rho_x = |\psi_x\rangle\langle\psi_x|.}$$

The unnormalized vector $M_x|\psi\rangle$ contains both pieces of information:

- its squared norm is the outcome probability,
- its direction gives the conditional pure state.

If $M_x|\psi\rangle = 0$, that outcome cannot occur.

A general outcome can involve several measurement operators and leave a mixed conditional state even for a pure input. We treat this case using quantum instruments in Section 4.5.

Consistency with ensemble preparations

The postulated update agrees with applying the pure-state rule to each preparation and conditioning on the observed outcome.

Write $\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$ and $q(x \mid j) = \|M_x|\psi_j\rangle\|^2$. For $p(x) > 0$, the joint probabilities and Bayes' rule give

$$p(j, x) = p_j q(x \mid j), \quad p(x) = \sum_j p_j q(x \mid j), \quad p(j \mid x) = \frac{p_j q(x \mid j)}{p(x)}.$$

Average the output states with these conditional weights:

$$\begin{aligned} \rho_x &= \sum_{j : q(x|j) > 0} \frac{p_j q(x \mid j)}{p(x)} \frac{M_x |\psi_j\rangle\langle\psi_j| M_x^\dagger}{q(x \mid j)} \\ &= \frac{\sum_j p_j M_x |\psi_j\rangle\langle\psi_j| M_x^\dagger}{p(x)} = \frac{M_x \rho M_x^\dagger}{p(x)}. \end{aligned}$$

The result depends only on the density operator ρ. □

Realizing a POVM

Given any POVM $\{\mu(x)\}_x$ on $\mathcal{H}$, choose

$$M_x = \sqrt{\mu(x)}.$$

The positive square root satisfies

$$M_x^\dagger M_x = \mu(x), \qquad \sum_x M_x^\dagger M_x = \sum_x \mu(x) = I.$$

This gives at least one measurement realization of the POVM. Other realizations can have different output states.

Projective measurements

For a projective measurement $\{P_x\}_x$, the choice $M_x = P_x$ gives the Lüders update

$$\boxed{\rho_x = \frac{P_x \rho P_x}{\mathrm{tr}[P_x \rho]}} \qquad \text{when } \mathrm{tr}[P_x \rho] > 0.$$

The same POVM, different output states

Consider two realizations of the computational-basis POVM:

$$M_0 = |0\rangle\langle 0|, \quad M_1 = |1\rangle\langle 1|,$$
$$N_0 = |0\rangle\langle 0|, \quad N_1 = |0\rangle\langle 1|.$$

For $x = 0, 1$, both give the same effect,

$$M_x^\dagger M_x = N_x^\dagger N_x = |x\rangle\langle x|,$$

and hence the same probability $p(x) = \langle x|\rho|x\rangle$. For outcomes with positive probability:

Outcome	Output using M_x	Output using N_x
0	$\|0\rangle\langle 0\|$	$\|0\rangle\langle 0\|$
1	$\|1\rangle\langle 1\|$	$\|0\rangle\langle 0\|$

The POVM alone does not determine the residual quantum state.

Discarding the measurement outcome

If we perform the measurement but discard its outcome, we average the conditional output states:

$$\begin{aligned}\rho' &= \sum_{x:p(x)>0} p(x)\rho_x \\ &= \sum_x M_x \rho M_x^\dagger.\end{aligned}$$

A zero-probability outcome contributes zero, since $M_x \rho M_x^\dagger \geq 0$ has trace zero.

A linear update

Write $\mathcal{E}(\rho) = \sum_x M_x \rho M_x^\dagger$. Then

$$\mathcal{E}(q\rho + (1-q)\sigma) = q\mathcal{E}(\rho) + (1-q)\mathcal{E}(\sigma)$$

for $0 \leq q \leq 1$. The completeness condition ensures $\mathrm{tr}[\mathcal{E}(\rho)] = 1$.

An unread computational-basis measurement

Let $P_0 = |0\rangle\langle 0|$, $P_1 = |1\rangle\langle 1|$, and write a qubit state as

$$\rho = \begin{pmatrix} a & b \\ \overline{b} & 1-a \end{pmatrix}.$$

The two unnormalized outputs are

$$P_0\rho P_0 = \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_1\rho P_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1-a \end{pmatrix}.$$

Discarding the outcome gives

$$\boxed{\rho' = P_0\rho P_0 + P_1\rho P_1 = \begin{pmatrix} a & 0 \\ 0 & 1-a \end{pmatrix}.}$$

The diagonal probabilities survive, while the off-diagonal entries vanish.

A pure state can become mixed

For input $|+\rangle\langle+|$, the unread computational-basis measurement gives

$$\rho = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \longmapsto \quad \rho' = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I}{2}.$$

Keeping the outcome

The pure outputs $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ each occur with probability $1/2$.

Discarding the outcome

The average $I/2$ is mixed. Unitary conjugation preserves purity.

Conclusion

Unitary evolution does not describe every physical operation on a quantum system. Performing a measurement and discarding its outcome gives a linear operation on density operators that need not be unitary.

Successive measurements

First measure using $L_l \colon \mathcal{H} \to \mathcal{K}$, then using $M_m \colon \mathcal{K} \to \mathcal{L}$, where

$$\sum_l L_l^\dagger L_l = I_{\mathcal{H}}, \qquad \sum_m M_m^\dagger M_m = I_{\mathcal{K}}.$$

Record both outcomes (l, m). Define

$$\boxed{N_{lm} = M_m L_l.}$$

These operators satisfy the completeness condition:

$$\begin{aligned} \sum_{l,m} N_{lm}^\dagger N_{lm} &= \sum_l L_l^\dagger \left(\sum_m M_m^\dagger M_m \right) L_l \\ &= \sum_l L_l^\dagger L_l = I_{\mathcal{H}}. \end{aligned}$$

We now check that this single measurement reproduces both the joint probabilities and the conditional states.

The joint outcome probabilities

The probability of the first outcome is

$$p(l) = \mathrm{tr}[L_l \rho L_l^\dagger].$$

If $p(l) > 0$, the first conditional state is $\rho_l = L_l \rho L_l^\dagger / p(l)$. Therefore

$$\begin{aligned} p(m \mid l) &= \mathrm{tr}[M_m \rho_l M_m^\dagger] \\ &= \frac{\mathrm{tr}[M_m L_l \rho L_l^\dagger M_m^\dagger]}{p(l)}. \end{aligned}$$

Multiplication by $p(l)$ cancels the first normalization:

$$\boxed{p(l, m) = \mathrm{tr}[M_m L_l \rho L_l^\dagger M_m^\dagger] = \mathrm{tr}[N_{lm} \rho N_{lm}^\dagger].}$$

If $p(l) = 0$, positivity implies $L_l \rho L_l^\dagger = 0$, so the same formula gives $p(l, m) = 0$.

The final conditional state

Suppose $p(l, m) > 0$. After the second measurement, the state is

$$\begin{aligned}\rho_{lm} &= \frac{M_m \rho_l M_m^\dagger}{p(m \mid l)} \\ &= \frac{M_m L_l \rho L_l^\dagger M_m^\dagger}{p(l)\, p(m \mid l)} \\ &= \boxed{\frac{N_{lm} \rho N_{lm}^\dagger}{p(l, m)}.}\end{aligned}$$

Thus the operators $N_{lm} = M_m L_l$ reproduce the full measurement procedure. □

Order matters

The first operator acts on the right: first L_l, then M_m.

The conclusions so far

Closed-system evolution

$$\rho \longmapsto U\rho U^\dagger, \qquad U(t) = e^{-itH/\hbar}, \qquad \dot{\rho} = -\frac{i}{\hbar}[H, \rho].$$

Unitary evolution preserves the spectrum and purity.

Measurement with a recorded outcome

$$p(x) = \mathrm{tr}[M_x \rho M_x^\dagger], \qquad \rho_x = \frac{M_x \rho M_x^\dagger}{p(x)} \quad (p(x) > 0).$$

The POVM $\mu(x) = M_x^\dagger M_x$ specifies probabilities, but does not determine the output states.

Measurement with a discarded outcome

$$\rho \longmapsto \sum_x M_x \rho M_x^\dagger.$$

This linear update can turn a pure state into a mixed state.